

Vikas Patel Pulkam

Data Engineer (AI & ML)

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PROFESSIONAL SUMMARY

Data Engineer (AI & ML) with **7+ years** of experience designing, building, and optimizing large-scale data platforms, feature stores, and MLOps pipelines across healthcare, entertainment, and e-commerce industries. Proven success in developing real-time and batch data systems leveraging **AWS, Azure, Databricks, and Apache Spark** to enable scalable, low-latency machine learning workflows. Proficient in **Python** and **SQL** for data transformation, automation, and analytics. Skilled in **ETL/ELT design, feature engineering, data lake architecture, and CI/CD automation** using modern tools such as **dbt, Kafka, MLflow, Terraform, and Airflow**. Adept at collaborating with cross-functional teams to ensure **data quality, pipeline reliability, and model performance**.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Bash

Big Data & Distributed Computing: Apache Spark (PySpark, Spark SQL), Hadoop (HDFS, YARN, MapReduce, Hive, Pig)

Data Processing & Orchestration: Apache Airflow, dbt (Data Build Tool), AWS Glue, Azure Data Factory, Databricks, SSIS

Streaming & Real-Time Systems: Apache Kafka, Spark Structured Streaming, AWS Kinesis, Apache Flink

ML & Feature Engineering: TensorFlow, PyTorch, scikit-learn, XGBoost, pandas, NumPy, PySpark MLlib, Feature Store Design

MLOps & Model Deployment: AWS SageMaker, Azure ML, TensorFlow Serving, MLflow, Kubeflow

DevOps & Infrastructure Automation: Docker, Kubernetes, Jenkins, Terraform, GitHub Actions, Git, CI/CD Pipelines, IaC

Cloud Platforms: AWS (S3, EMR, Redshift, Glue, Lambda), Azure (Data Lake Storage, Synapse, Databricks, Azure ML), Snowflake

Databases & Data Warehousing: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, DynamoDB

Methodologies & Practices: Agile (Scrum, SAFe), DataOps, Data Lake Architecture, Data Modeling, Cloud Cost Optimization

Certifications: AWS Certified Data Engineer - Associate

PROFESSIONAL EXPERIENCE

Netflix | CA | Data Engineer (AI & ML)

March 2025 – Present

- Designed and implemented the core architecture for a **low-latency feature platform** leveraging **AWS S3, DynamoDB, and Apache Flink**, enabling millisecond-latency feature computation and serving for **hundreds of production ML models**.
- Optimized **petabyte-scale batch feature engineering pipelines** using **Apache Spark (PySpark)** on Netflix's internal data platform (**Genie**), improving processing performance by **20%** through advanced partitioning and caching strategies.
- Built and deployed a **high-throughput gRPC/HTTP feature-serving API** integrated with **Apache Kafka**, unifying real-time (**DynamoDB**) and batch (**S3**) feature stores and reducing online inference latency from **100ms to under 25ms**.
- Engineered a **scalable dbt Core-based transformation layer** to model and version batch features, introducing automated **data quality testing** that improved feature reliability by **15%** for **ML training workflows**.
- Developed **Python-based utilities** (NumPy, pandas, scikit-learn) for data preprocessing, feature validation, and **automated drift detection**, enhancing production model stability and accuracy by **7%**.
- Partnered cross-functionally with **Machine Learning Engineers, Data Scientists, and Infrastructure teams** to define feature **SLAs, streamline data onboarding**, and enable faster **ML model integration**, accelerating deployment cycles by **20%**.

Optum RX/Optum AI | IL | Senior Data Engineer

October 2023 – February 2025

- Architected a **scalable healthcare data lake** on **Azure Data Lake Storage Gen2, Azure Data Factory, and Azure Databricks**, centralizing **10TB+ of structured and unstructured clinical, claims, and pharmacy data daily**.
- Optimized **enterprise-scale ETL/ELT workflows** using **PySpark** with incremental loading and parallel processing on **Databricks**, reducing pipeline execution time by **25%** and **cloud infrastructure costs** by **15%**.
- Built **real-time ML monitoring dashboards** leveraging **Spark Streaming, Tableau, and Python (Plotly, Dash)** to visualize model drift and prediction accuracy, improving data-driven decision latency by **10%**.
- Automated **end-to-end MLOps pipelines** using **MLflow, Jenkins, and Terraform**, improving production **ML model deployment** reliability and governance by **20%** across batch and streaming workflows.
- Implemented **SQL-driven hyperparameter tuning** via **Snowflake UDFs**, orchestrated feature preparation with **dbt**, improving model accuracy by **8%** while reducing training cost by **12%** through efficient parameter search.
- Led **Agile/Scrum ceremonies** and mentored **3 junior engineers** in sprint planning, backlog refinement, and **dbt best practices**, increasing overall team delivery velocity by **20%**.

- Designed and implemented a **multi-zone data lake** on **Amazon S3** following the **medallion architecture** (Raw, Enriched, Curated), processing **1.5+ TB of daily clickstream data** using **AWS Glue** and **EMR** to establish a single source of truth for advertising attribution.
- Architected and optimized **ETL/ELT pipelines** using **AWS Glue**, **PySpark**, and **SQL**, transforming raw clickstream data into enriched datasets, improving data processing efficiency by **15%** and reducing analytics time-to-insight.
- Built **real-time streaming pipelines** leveraging **Apache Kafka** and **AWS Kinesis** to handle high-velocity clickstream ingestion, reducing event processing latency from **minutes to under 10 seconds** and enabling faster ad bidding decisions.
- Created and managed a **centralized feature store** using **Amazon SageMaker Feature Store**, computing and serving thousands of customer-level features for ML models predicting ad conversion likelihood.
- Developed **Python-based automation tools** (boto3, pandas) for **data quality validation**, **cloud resource management**, and **report generation**, reducing manual operational work and improving data reliability by **10%**.
- Engineered **CI/CD pipelines** using **AWS CodePipeline** and **GitHub Actions** to automate deployment of **Glue jobs**, **Airflow DAGs**, and **SQL transformations**, cutting manual deployment errors by **20%** and increasing release consistency.

Mu Sigma Inc. | India | Data Engineer

March 2017 – March 2019

- Implemented **automated data quality checks** in **Python** and scheduled validation jobs using **Apache Airflow**, ensuring ingestion accuracy and reducing data quality incident reports by **15%**.
- Designed and deployed **ETL pipelines** using **Apache Spark (PySpark)** to process **terabytes of daily retail transaction data**, integrating multiple data sources and reducing manual processing time by **20%**.
- Engineered and optimized **large-scale batch processing workflows** within the **Hadoop ecosystem (Hive, Pig, MapReduce)** to transform and aggregate historical datasets, improving batch processing efficiency by **12%** for quarterly business analytics.
- Created **reusable SQL templates** and **process documentation** to enable **client analytics teams** to independently query **Oracle** and the new data platform, accelerating time-to-insight and reducing dependency on engineering teams.
- Monitored and optimized **AWS Redshift** cluster performance and **S3 storage lifecycle**, applying cloud best practices that reduced monthly data infrastructure costs by **5%**.
- Collaborated with **business analysts**, **data scientists**, and **client stakeholders** to translate business requirements into scalable data solutions, improving cross-functional alignment and reducing requirement gaps by **10%**.

EDUCATION

Master's in Information studies | Trine University, Detroit, Michigan, USA.**Bachelor's in Electrical & Electronics Engineering** | Guru Nanak Institute of Technology, Hyderabad, India.